

Amjad S Al-thabteh

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Education

B.S in Computer Science	GPA: 3.55/4.00	<i>Wayne State University</i>	<i>Detroit</i>	<i>May</i>
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Experience

<i>SWE Intern</i>	<i>Immersible Graphics</i>	<i>Detroit, MI</i>	<i>Sep 2025</i>
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- Architected a real-time rendering engine in **C++20 and OpenGL**, focusing on **GPU-driven rendering**.
- Robust internal UI and asset-authoring tools built on JSON-driven pipelines, greatly expanding the flexibility of content creation.
- **Data-driven rendering optimizations** to allow runtime analysis and AI guided heuristics to balance visual quality.
- Modular rendering subsystems with clear ownership boundaries to support iteration and feature growth

<i>Software Development Intern</i>	<i>Rare Munchies</i>	<i>Ann Arbor, MI</i>	<i>Sep 2023 - Apr 2024</i>
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- Integrated UI components using TypeScript, improving page transitions and interaction flow to enhance user experience.
- Resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- **Cross-browser compatibility issues**, reducing debugging overhead and stabilizing releases

Projects

Glint 3D – C++ Automated 3D Rendering Engine

[Glint Render](#)

- Rendering system built on **C++17/OpenGL rendering pipeline with dynamic GPU workload** and lighting optimization.
- Designed a cross-platform architecture with runtime scene analysis to maintain stable performance in complex scenes
- Designed a modular engine architecture separating rendering, scene management enabling faster iterations.
- Implemented a custom memory pool and allocator system using C++20, significantly reducing heap allocations.

UE -5.6 Cinematic “No One Answers Clip”

[No one answers clip](#)

- Built a solo AAA-quality cinematic in Unreal Engine 5.6 using Metahuman animation, cinematic cameras, control systems.
- Used Niagara effects and environmental animation to tell the story visually and set the tone for No One Answers
- Created as a vertical-slice cinematic for an in-development narrative game.

High-Frequency Trading Engine (C++20)

- Engineered a low-latency limit order book with FIFO matching logic, spread crossing detection, and trade aggregation, leveraging modern **C++20** features and cache-efficient containers to model high-frequency execution flow.
- Implemented performance benchmarking and latency measurement tools, (microsecond/nanosecond precision) to analyze algorithms.

C++ Runtime Call Stack & Diagnostics Engine (Trace forge)

- Developed a **C++ 23 debugging system** that performs automated stack-trace parsing.
- Implemented automated regression test generation and rule-based fix suggestions derived from stack trace patterns
- Added **thread-safe logging** to capture errors and runtime events in multithreaded applications.
- Basic **memory tracking utilities** to help identify leaks and invalid allocations during debugging.

UE 5.5 Game

- Wrote gameplay systems **using (C++ 20 and Visual Scripting)**, connecting animation states, environmental triggers.
- Created 3d environments and characters from scratch, rigging for animation, and enhanced each scene.
- Produced a 90+ second real-time cinematic using Metahumans, Control Rig, and Sequencer designing the flow.

Skill

- **Programming Languages:** C++ 11 - 26, C#, C, Python 3.7 - 3.12, TypeScript, Java
- **Graphics & Engines:** OpenGL Unreal Engine 4 - 5.7 • Unity • • NX Design
- **Course Work:** Computer Graphics, Game Programming, **Software Engineering, Operating Systems**, Linear Algebra
- **Specialties:** Gameplay Systems, VR/AR Development, Simulation, Tools & Plugins, Technical Art